Assignment 1

1️⃣ Zero-Shot Prompting

Zero-shot prompting means giving the model a task **without providing any examples**.
You simply describe what you want, and the model performs the task using its prior training.

- **Example**

Prompt: "Explain Machine Learning in simple terms."

Here, you didn't provide any sample explanation — the model generates the answer directly.

2️⃣ One-Shot Prompting

One-shot prompting means giving **one example** of how the output should look before asking the model to perform the task.

- **Example**

Prompt: Translate English to French.
Example:
English: Hello
French: Bonjour

English: Thank you

Now the model understands the format from one example.

3️⃣ Few-Shot Prompting

Few-shot prompting means giving **multiple examples (2–5 or more)** before asking the model to perform the task.

- **Example**

Prompt: Classify sentiment:

Review: I love this product → Positive
Review: This is terrible → Negative
Review: It's okay, not great → Neutral

Review: The quality is amazing → ?

🔥 Quick Comparison Table

Feature	Zero-Shot	One-Shot	Few-Shot
Examples Provided	0	1	2+
Accuracy	Medium	High	Very High
Token Usage	Low	Medium	High
Best For	Simple tasks	Structured tasks	Complex pattern tasks

Solve this math problem: If a train travels 60 miles in 2 hours, what is its average speed?

To find **average speed**, use the formula:

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

Given:

- Distance = 60 miles
- Time = 2 hours

$$\text{Average Speed} = \frac{60}{2} = 30$$

✅ Final Answer:

The average speed of the train is 30 miles per hour (mph).

Solve these math problems. Example: Problem: A car travels 100 miles in 4 hours. What is its average speed? Answer: 25 miles per hour. Now solve: Problem: If a train travels 60 miles in 2 hours, what is its average speed?

Answer: 30 miles per hour.

Solve these math problems. Example 1: Problem: A car travels 100 miles in 4 hours. What is its average speed? Answer: 25 miles per hour. Example 2: Problem: A cyclist covers 45 miles in 3 hours. What is the average speed? Answer: 15 miles per hour. Example 3: Problem: A runner completes 10 miles in 2 hours. What is the average speed? Answer: 5 miles per hour. Now solve: Problem: If a train travels 60 miles in 2 hours, what is its average speed?

Answer: 30 miles per hour.